Bull-CNN: Next-Generation News Search & Stance-Aware Perspective Recommender



OUR MISSION TODAY: FROM VECTOR DILUTION & ECHO CHAMBERS TO GLASS-BOX NEWS INTELLIGENCE

Built from mathematical first principles on 37,916 CNN articles • Verified on 250 controlled benchmark queries • Ready for sovereign BullSequana AI deployment

EXECUTIVE AGENDA • PRESENTATION STRUCTURE

Agenda: From Scientific PoC to Live Demo & 3-Month Rollout

Three core goals on the left connected to their four milestone building blocks inside each circle



GOAL 1 • PART I (A)

Exploratory Data Analysis & Data Cleaning Pipeline

Corpus Overview:
Dataset Scale, Timeline & 55 Sections

Token Length Analysis:
BPE Token Lengths & Long-Article Tail

Preprocessing Pipeline:
Data Cleaning & Indexing Funnel

GOAL 1 • EXPLORATORY DATA ANALYSIS — PART 1: CORPUS OVERVIEW

CNN Corpus Overview: 11 Years, 9 Categories & 55 Sections

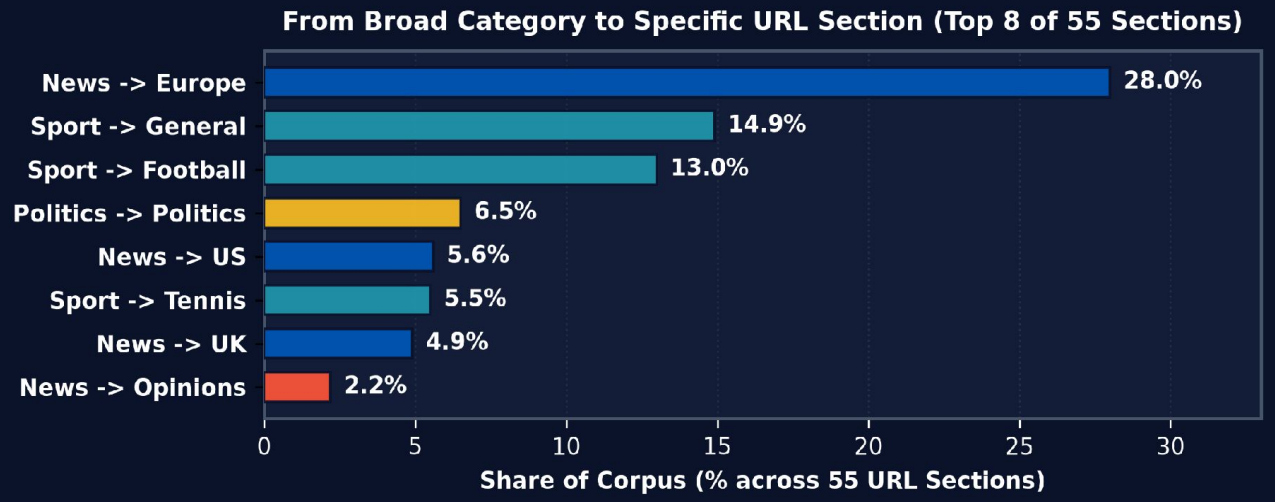
Analyzing dataset scale, publication timeline, and editorial taxonomy across 37,916 articles

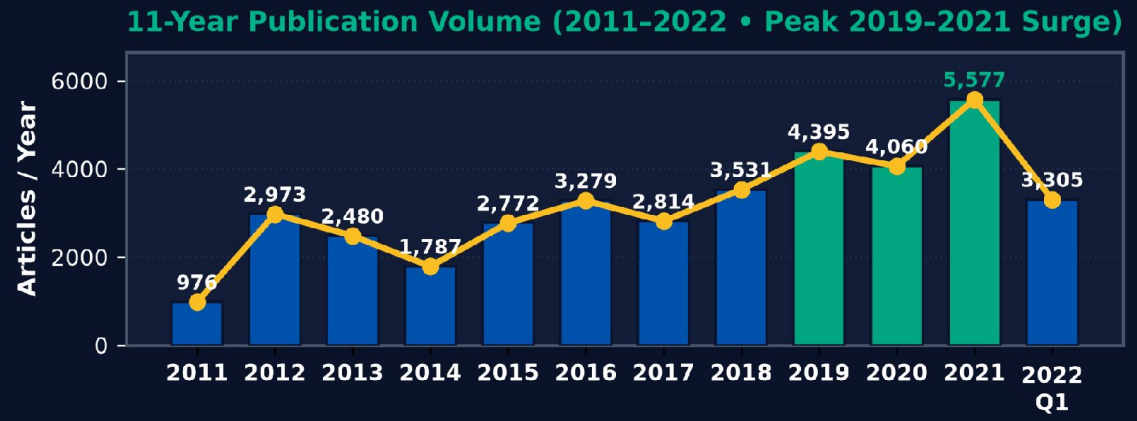
1. Dataset Scale & 11-Year Timeline

- 37,916 clean CNN articles spanning March 2011 to Ma 2022
- Publication volume surges between 2019 and 2021, peaking at 5,577 articles in 2021

2. Taxonomy: 9 Categories & 55 Sections

- Top-level Categories are led by News at 47.6% and Sp at 41.0%, followed by Politics at 6.5%, Business, Healt Entertainment
- Parsing URL paths unlocks 55 fine-grained editorial de led by Europe at 28.0%, Sport at 14.9%, Football at 13 Politics, US, Tennis, UK & Opinions





Exploratory Data Analysis: Corpus & BPE Token Lengths

1. Shared BPE Subword Tokenizer

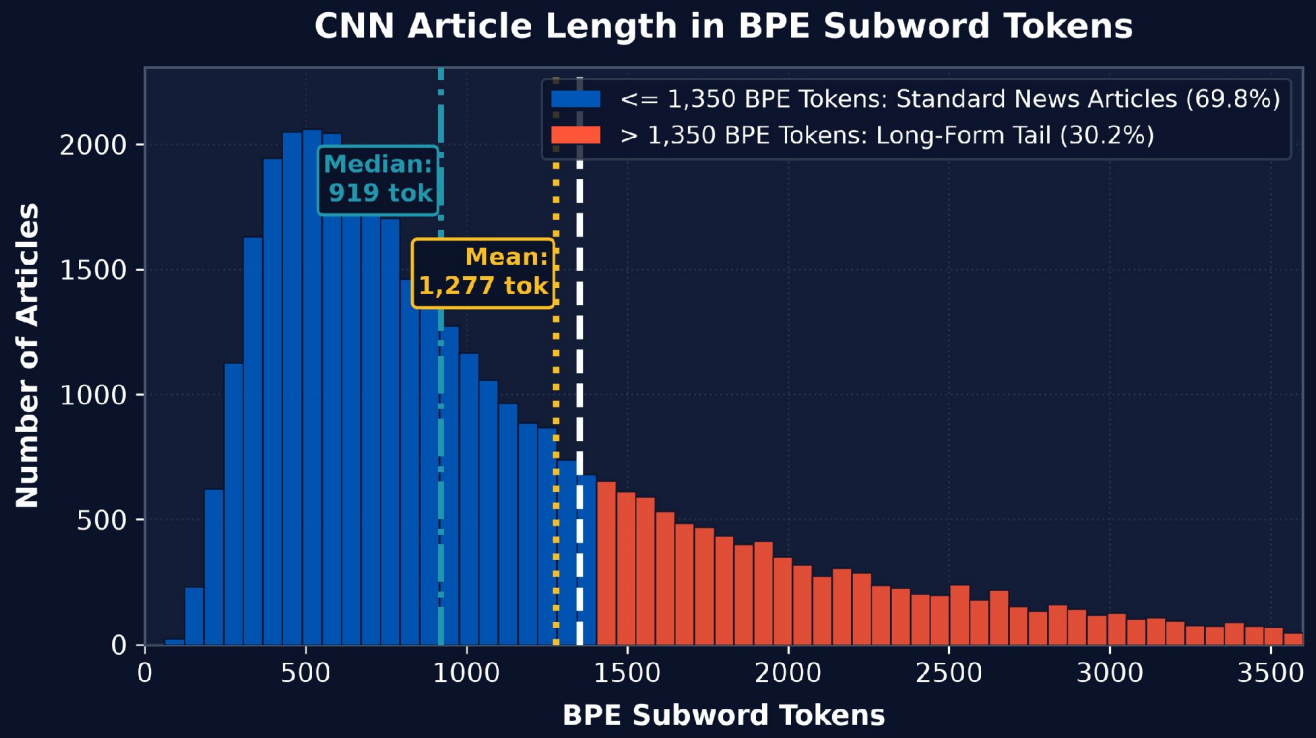
- Byte-Pair Encoding splits words into subword tokens
- Used models: Harrier-0.6B, Qwen3-Reranker & Qwen3.8
- all share the identical 151,643-token BPE vocabulary

2. Token Length Distribution & Long-Form Tail

- Headlines have a median of 14 tokens and summaries have a median of 33 tokens
- Full articles have a median of 919 tokens and a mean of 1,277 tokens
- 46.0% exceed 1,000 BPE tokens, and 30.2% exceed 1,350 tokens up to 23,670 BPE tokens

3. HTML Paragraph Flattening

- 77.2% of articles have zero newline breaks because the scraper flattened HTML paragraph tags into one block



💡 KEY FINDING: WHY NAIVE CHUNKING & WHOLE-DOCUMENT VECTORS FAIL ON THIS CORPUS

Because 77.2% of articles lack paragraph breaks and 30.2% span 1,350 to 23,670 BPE tokens, we cannot split on paragraphs or compress long articles into a single vector without diluting buried facts — motivating fixed BPE token windows with context headers and length-debiased scoring.

GOAL 1 • DATA CLEANING & PREPARATION

Data Cleaning Funnel: From Raw Scrape to 37,916 Clean Articles

Removing newsletter ads, empty stubs, and duplicate headlines in plain, auditable steps

RAW WEB SCRAPE

37,949 Uncleaned Articles

Newsletter & Promo Ads

"Sign up for CNN Meanwhile in China..." mixed into story text

33 Empty Articles

Missing body text or only a 1-line reporter credit

175 Duplicate Headlines

Recurring weekly quiz & live-update titles

Raw URLs & Title Suffixes

"- CNN" in titles & unparsed category links

STEP 1 CLEAN TEXT

- Strip newsletter ads & promo banners
- Remove '(CNN) --' wire headers
- Fix glued 'Story highlights' text

STEP 2 PARSE TOPICS

- Strip '- CNN' from headlines
- Extract 9 Categories & 55 Sections from URL
- Parse publication dates (2011–2022)

STEP 3 FILTER & DEDUP

- Remove 33 empty articles
- Mark 175 duplicate headlines
- Keep all 37,916 valid articles

CLEAN SEARCH CORPUS

37,916 Ready Articles

Complete Articles

33 empty stubs removed;
100% verified text, dates & sections

Deduplicated Headlines

175 recurring headlines marked so search never shows duplicates

3 Separate Search Fields

Clean Headline, Summary & Body ready for weighted search

GOAL 1 • PART I (B)

Evaluation Methodology & Hybrid Retrieval Architecture

Evaluation Paradigm:
Benchmark Design & LLM Judge

Hybrid Search:
BM25F, Dense MaxP & Reranking

Perspective Engine:
Stance Recommender & Leaderboard

GOAL 1 • EVALUATION-FIRST DRIVEN DEVELOPMENT

Benchmark Setup: The 5 Query Categories We Optimize For

Controlled multi-query benchmark across 50 anchor articles and 1,200 independent LLM relevance judgments

1. Controlled Multi-Query Benchmark

- 50 anchor articles × 5 query categories = 250 paired benchmark queries
- Anchor Example: "Zelensky pressed Biden for more sanctions during their latest call"

2. Independent Judge: Gemini 3.8 Flash Next

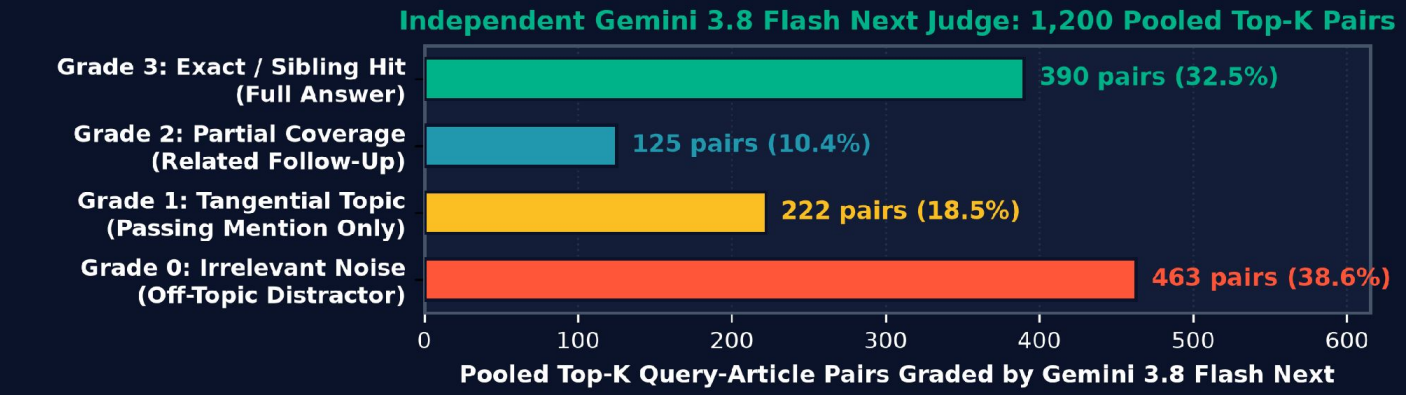
- We use Gemini 3.8 Flash Next as an independent Judge to grade 1,200 query-article pairs on a 0–3 relevance scale
- Overcomes LLM self-preference bias when evaluating our local Qwen reranker

3. Strict vs. Fair Sibling Metrics

- Strict Recall@5 requires the exact source article; Fair nDCG@10 also credits valid breaking-news follow-up articles

BENCHMARK EXAMPLE: 1 ANCHOR ARTICLE -> 5 QUERY CATEGORIES

PART OF ANCHOR ARTICLE	DERIVED BENCHMARK QUERY
Named Entity in Text "...Russian Direct Investment Fund"	**1. Keyword: "Russian Direct Investment Fund"** Exact named entity -> Solved by BM25F
Headline & Date: March 2022 "Zelensky pressed Biden... latest call"	**2. Temporal: "Biden Zelensky call March 2022"** Date + headline entities -> Solved by BM25F
Middle Body Passage: Chunk #3 "...punish Moscow... severe costs"	**3. Paraphrase: "harsh economic penalties on Moscow"** Zero shared words -> Solved by Debiased MaxP
Final Paragraph: Chunk #4 "...release 60M barrels of oil"	**4. Deep Fact: "How many barrels of oil released?"** Buried in Chunk #4 -> Solved by Passage Forwarding
Editorial Argument & Stance "...ratcheting up economic pressure"	**5. Stance: "supporting trade penalties on Russia"** Directional opinion -> Solved by Stance & Aspect-MMR



Step 1: Fielded BM25F — Strong on Keywords, Fails on Paraphrase

1. Solves Keyword & Temporal Lookups

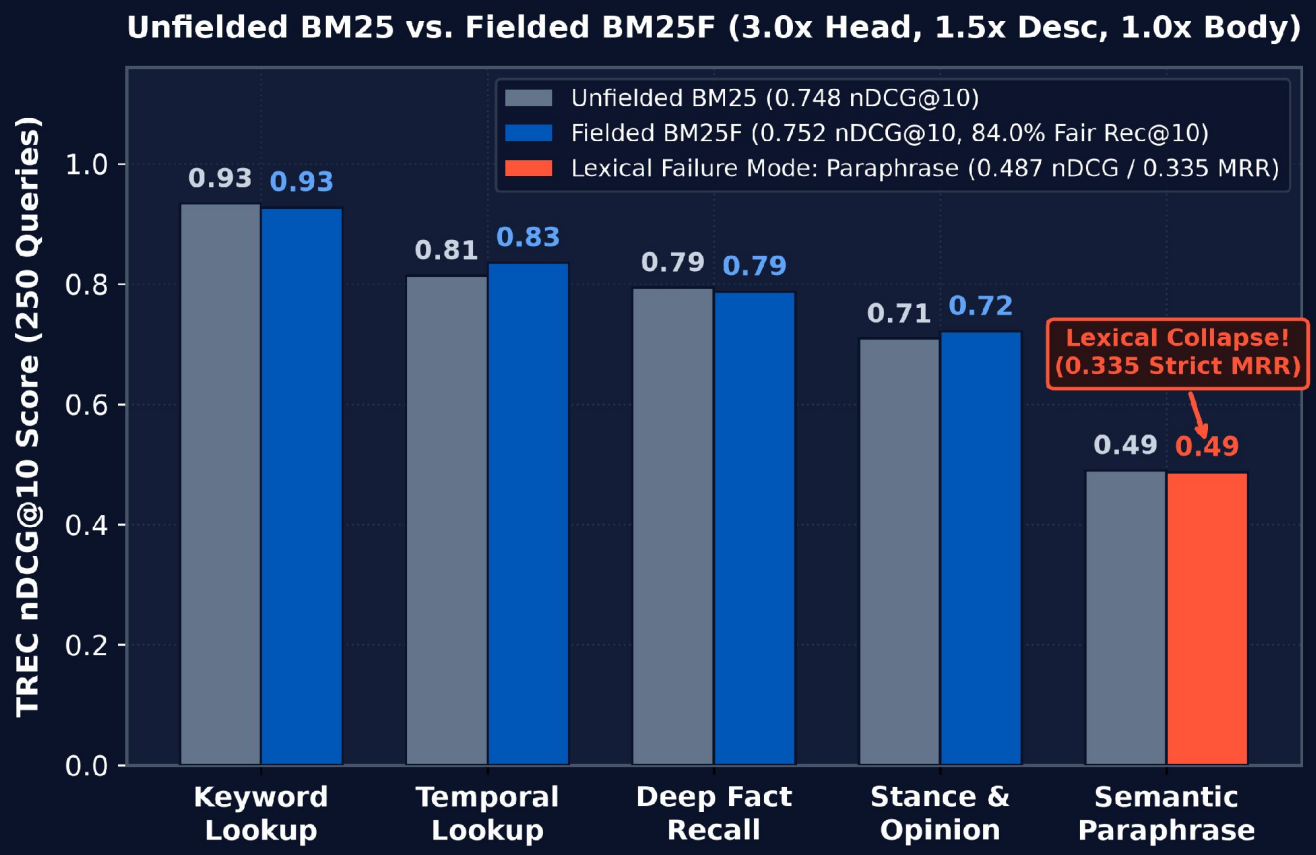
- Anchor Article: "Zelensky pressed Biden for more sanctions during their latest call"
- Weighting Headline 3.0x, Summary 1.5x, and Body 1.0x reaches 0.93 nDCG on Keyword Lookup

2. Fails on Semantic Paraphrase: 0.49 nDCG

- Paraphrase Query: "demanding harsh economic penalties on Moscow"
- Shares zero content words with the headline, causing BM25F to drop to 0.487 nDCG

3. Next Step: Add Dense Passage Vectors

- Keep BM25F as exact-entity anchor and add neural passage embeddings to resolve semantic paraphrases



Step 2: Contextual BPE Chunks & Length-Debiased MaxP

1. Why Single-Vector Article Averaging Fails

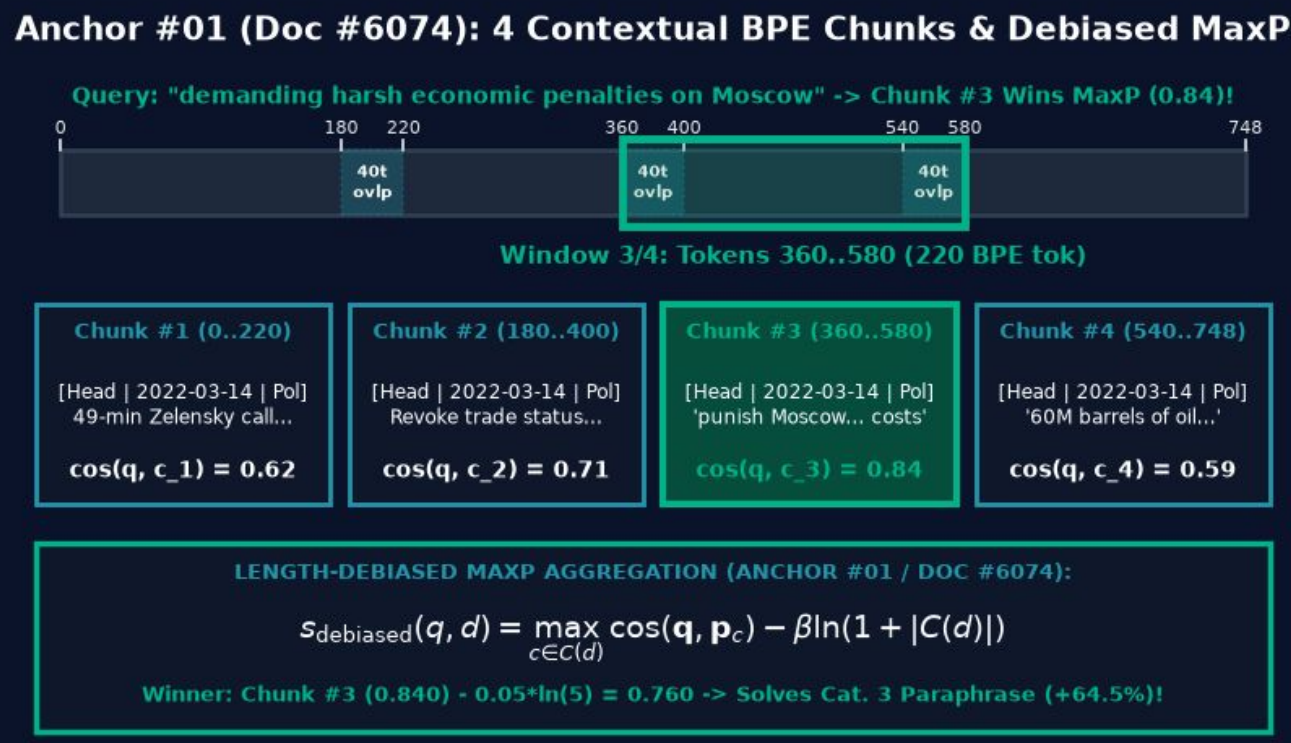
- We encode passages with Harrier-0.6B neural embeddings to capture semantic meaning beyond exact words
- Averaging a multi-chunk article into a single vector blurs specific paragraphs and drops nDCG to 0.678

2. Context-Stamped 220-Token BPE Windows

- Articles are sliced into 220-token BPE windows with a 40-token overlap, creating 208,187 passages across the corpus
- Every chunk is prefixed with Headline, Date, and Section so interior passages never lose document context

3. Length-Debiased MaxP Boosts Paraphrase by +64.5%

- Interior Chunk #3 matches "harsh economic penalties on Moscow" with 0.84 similarity
- Subtracting a logarithmic length penalty prevents long articles from dominating and lifts Paraphrase nDCG from 0.487 to 0.801



Step 3: Dynamic Alpha Hybrid — Fusing BM25F + Dense MaxP

Adapting lexical vs. semantic fusion weights per query using token rarity and score specificity

1. Why Static 50/50 Weights Are Sub-Optimal

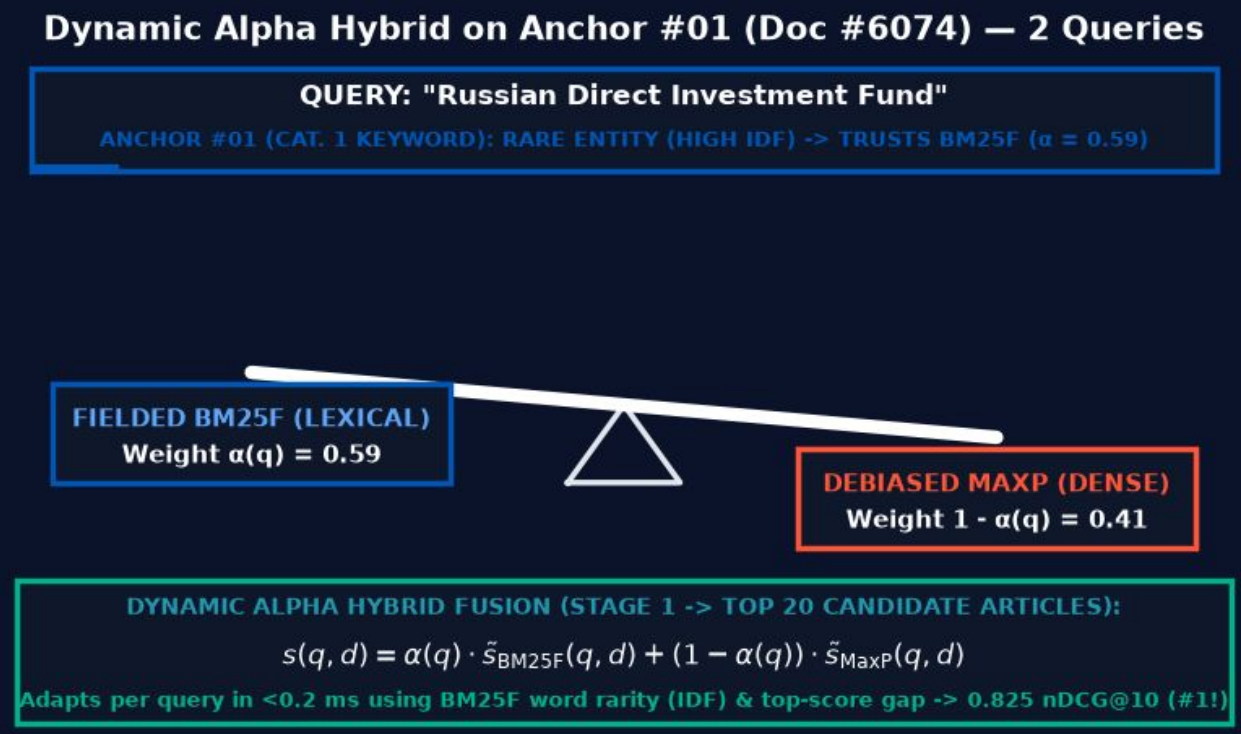
- BM25F excels on exact keywords and dates, while Dense Debiased MaxP excels on semantic paraphrases
- A fixed 50/50 split dilutes rare named entities on keyword queries and pulls down semantic paraphrases

2. Query-Adaptive Weighting in Under 0.2 ms

- Computes query token rarity and top-score separation to set lexical weight alpha dynamically:
- Keyword Query "Russian Direct Investment Fund" -> rare proper nouns set alpha = 0.59 to prioritize BM25F
- Paraphrase Query "harsh economic penalties on Moscow" -> common words set alpha = 0.26 to prioritize Dense MaxP

3. Fast Stage-1 Candidate Generation

- Retrieves the Top 20 candidate articles in 8.8 ms with 0.825 overall nDCG@10 across all 250 benchmark queries



Step 4: Two-Stage Funnel & Winning Passage Forwarding

PART 1: HOW ARTICLES AND PASSAGES (CHUNKS) FLOW THROUGH OUR TWO-STAGE SEARCH PIPELINE

FULL CORPUS INDEX
37,916 Articles = 208,187 Passages

- Each article is split into ~5.5 passages (220 BPE tokens each)
- Problem: Running the Cross-Encoder on all 208,187 passages would take ~52 minutes per query!

Stage 1: BM25F + Vector MaxP

STAGE 1 OUTPUT (<9 ms)
Top 20 Candidate Articles

- Stage 1 (BM25F + Vector MaxP Search) scores all 208,187 passages and groups them by article
- Returns the 20 best ARTICLES — and remembers which 1 passage (Chunk #m) won inside each article!

Forwards 1 Passage per Article

STAGE 2 RERANKER (44.6 ms)
Final Top 5 Articles

- Qwen3-Reranker-0.6B runs only 20 calls (1 call per candidate article)
- Each call reads: [Headline + Summary + That Article's 1 Winning Passage #m]

PART 2: WHAT GETS FORWARDED FOR EACH OF THE 20 CANDIDATE ARTICLES?

Example: Doc #6074 (748 BPE Tokens = 4 Passages)

Cat. 4 Query: "How many barrels of oil did allies release?"

Passage #0 (Headline & Intro):
"Zelensky pressed Biden for more sanctions..." (No oil barrels!)

Passages #1 – #2 (Body):
"...trade status, G7 export controls & IMF package..."

★ Winning Passage #3 (Final Chunk #4!):
"...allies agreed to release 60 million barrels of oil from reserves."

✗ STANDARD RERANKING (Sends Headline + Intro Passage #0 for Each of the 20 Articles)

- Cross-Encoder only sees Passage #0 and is completely blind to Winning Passage #3!
- Misses the '60 million barrels' answer buried in the final chunk -> Deep Fact nDCG drops to 0.384

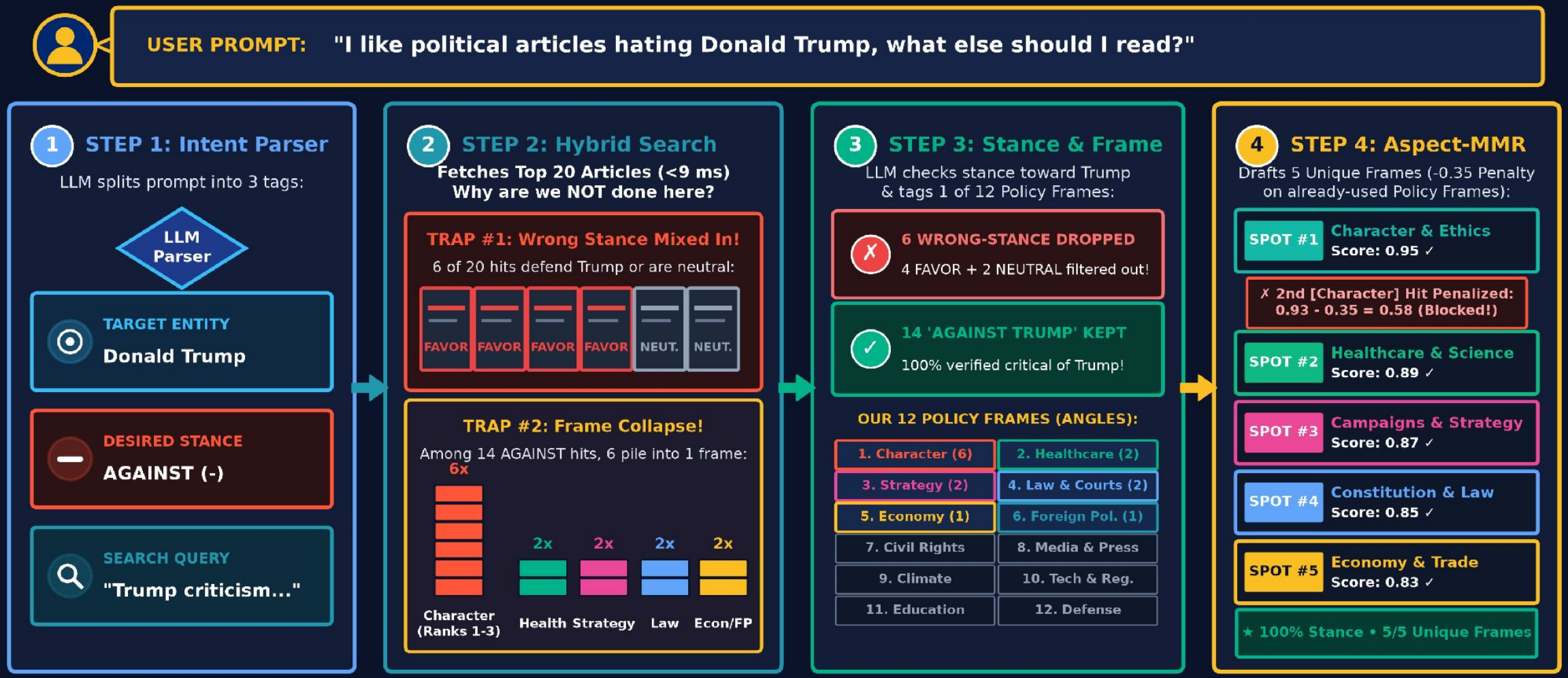
✓ OUR MAXP PASSAGE FORWARDING (Sends Headline + Summary + Winning Passage #3 per Article)

- **For each of the 20 Candidate Articles, we forward its 1 winning passage (#3) to Qwen3-Reranker-0.6B!**
- **Cross-Encoder reads '60 million barrels' directly -> Deep Fact nDCG nearly DOUBLES (0.384 -> 0.751)!**

GOAL 1 • STEP 5: TACKLING CATEGORY 5 — STANCE & PERSPECTIVE QUERIES

Step 5: Stance & 12-Frame Recommender — The 4-Step Pipeline

Visual building blocks: Speech Bubble -> LLM Intent Prism -> 20 Mini Article Cards -> Stance Filter & 12 Frames -> 5-Slot MMR Podium



Inside Step 4: How Aspect-MMR Fills the 5 Recommendation Spots

1. Starting Point from Step 3: 14 Verified 'AGAINST' Articles

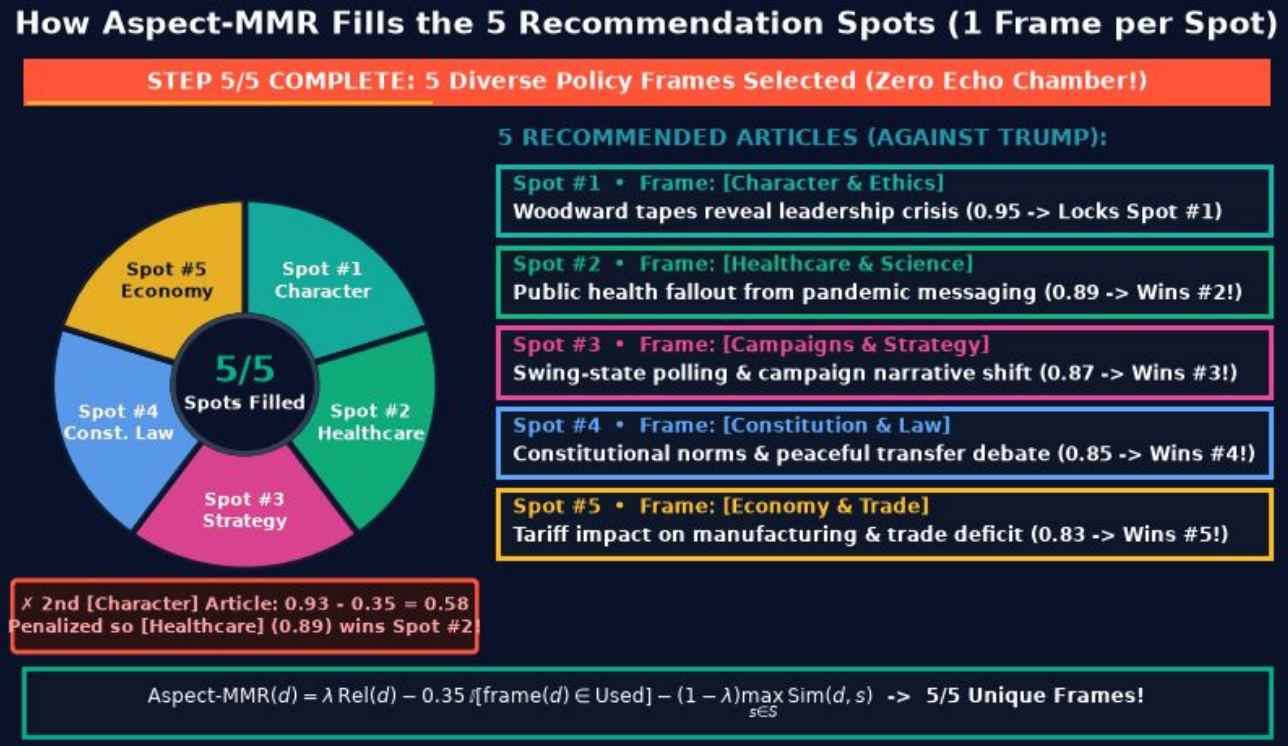
- Step 3 filtered the Top 20 search hits down to 14 articles that are 100% AGAINST Donald Trump, tagged with their Policy Frame (1 of 12)
- Problem: Ranks #1 (0.95), #2 (0.93), and #3 (0.91) all belong to Frame #1 [Character & Ethics]!

2. How Aspect-MMR Drafts the 5 Spots (-0.35 Penalty)

- Like filling 5 positions on a team, once a Policy Frame is in the basket, any further article from that same frame gets a -0.35 score penalty:
- Spot #1 takes [Character & Ethics] (0.95); the 2nd [Character] article drops (0.93 - 0.35 = 0.58 ✗), so [Healthcare] (0.89) wins Spot #2!

3. Result: 5/5 Unique Frames + Instant 1-Click Flip

- Delivers 100% matching stance across 5 distinct Policy Frames — plus 1-Click Flip (AGAINST <-> FAVOR) in <2 ms!

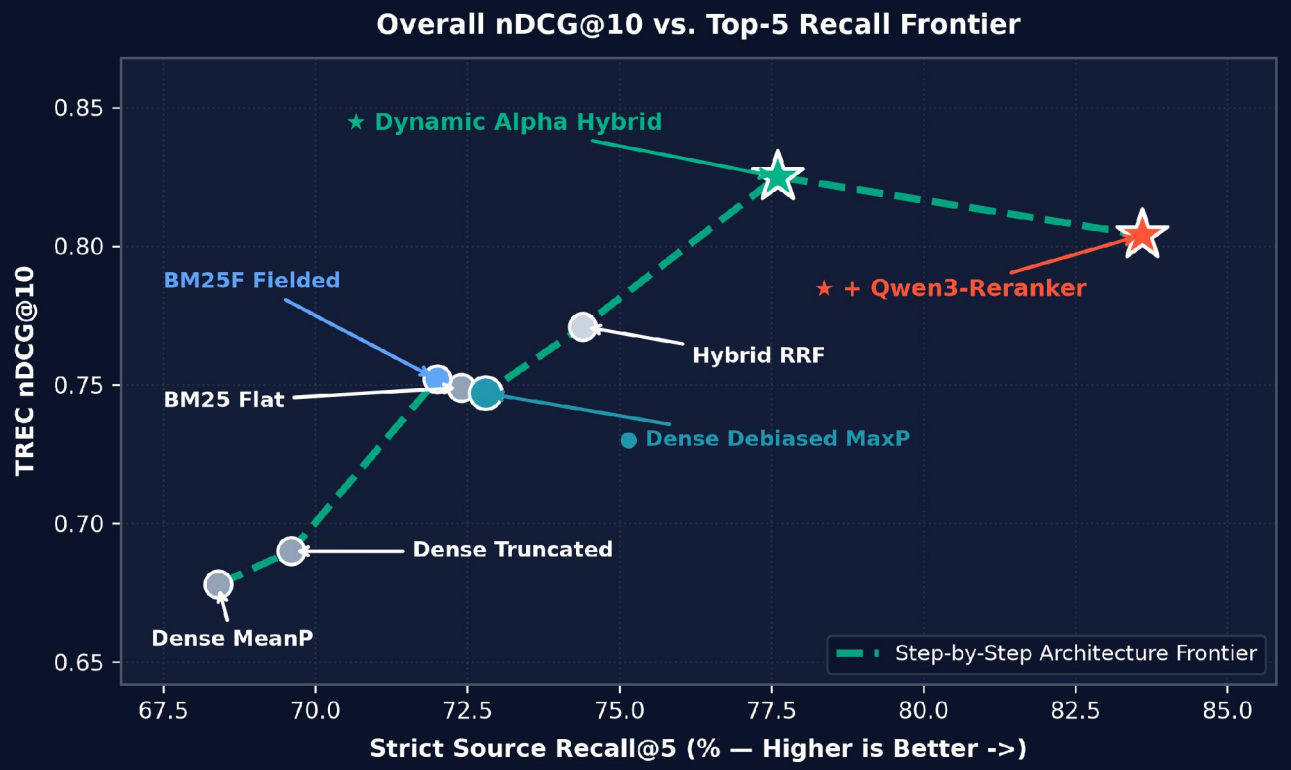


GOAL 1 • FULL-CORPUS BENCHMARK RESULTS (250 QUERIES, 37,916 ARTICLES, GEMINI 3.8 FLASH JUDGE)

Cumulative Leaderboard: 250-Query Pareto Frontier

Step-by-step upward progression across 250 queries and our 2 ★ production operating modes

System Architecture	nDCG	Fair MRR	Rec@5	Str MRR	Latency
BM25 Flat (Unfielded)	**0.749**	0.779	**72.4%**	0.619	12.6 ms
BM25F Fielded (3x/1.5x/1x)	**0.752**	0.780	**72.0%**	0.625	12.4 ms
Dense MeanP (Chunk Avg)	**0.678**	0.718	**68.4%**	0.562	7.9 ms
Dense Truncated (Lead Only)	**0.690**	0.752	**69.6%**	0.605	1.6 ms
• Dense Debiased MaxP	0.747	0.789	72.8%	0.619	8.2 ms
Hybrid RRF (Static 50/50)	**0.771**	0.793	**74.4%**	0.608	29.7 ms
★ Mode 1: Dynamic Alpha Hybrid	0.825	0.847	77.6%	0.659	8.8 ms
★ Mode 2: + Qwen3-Reranker	0.804	0.839	83.6%	0.677	44.6 ms



💡 **WHEN TO USE WHICH OPERATING MODE:**

★ **Use Mode 1 (Dynamic Alpha Hybrid)** when instant responses (latency <10ms) and broad top-10 ranking quality are required.

★ **Use Mode 2 (+ Qwen3 Cross-Encoder Reranker)** when maximum top-5 precision and deep body-fact recall are required.

GOAL 1 • ENGINEERING REFLECTION, LIMITATIONS & OUTLOOK

Current Limitations & Production Calibration Roadmap

Four high-leverage improvements when transitioning from our PoC to customer deployment



1. Human Gold Alignment (AI vs. Human)

Limitation: LLM-as-a-Judge can exhibit self-preference bias toward neural models.

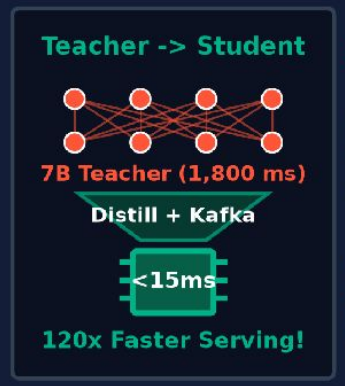
Outlook: Co-build a 200-query human gold set with customer editors & calibrate via Kendall's tau / Cohen's kappa.



2. Collaborative Click & Mouse Signals

Limitation: PoC relies purely on content & stance without historical user interaction logs.

Outlook: Fuse clickstream, dwell-time & skip signals via Team-Draft Interleaving & Counterfactual LTR.



3. Write-Time Enrichment & Distillation

Limitation: Uncached online LLM stance extraction takes ~1.8s on cold queries.

Outlook: Move stance/frame tagging to async Kafka ingest workers & distill into a <15ms encoder head.



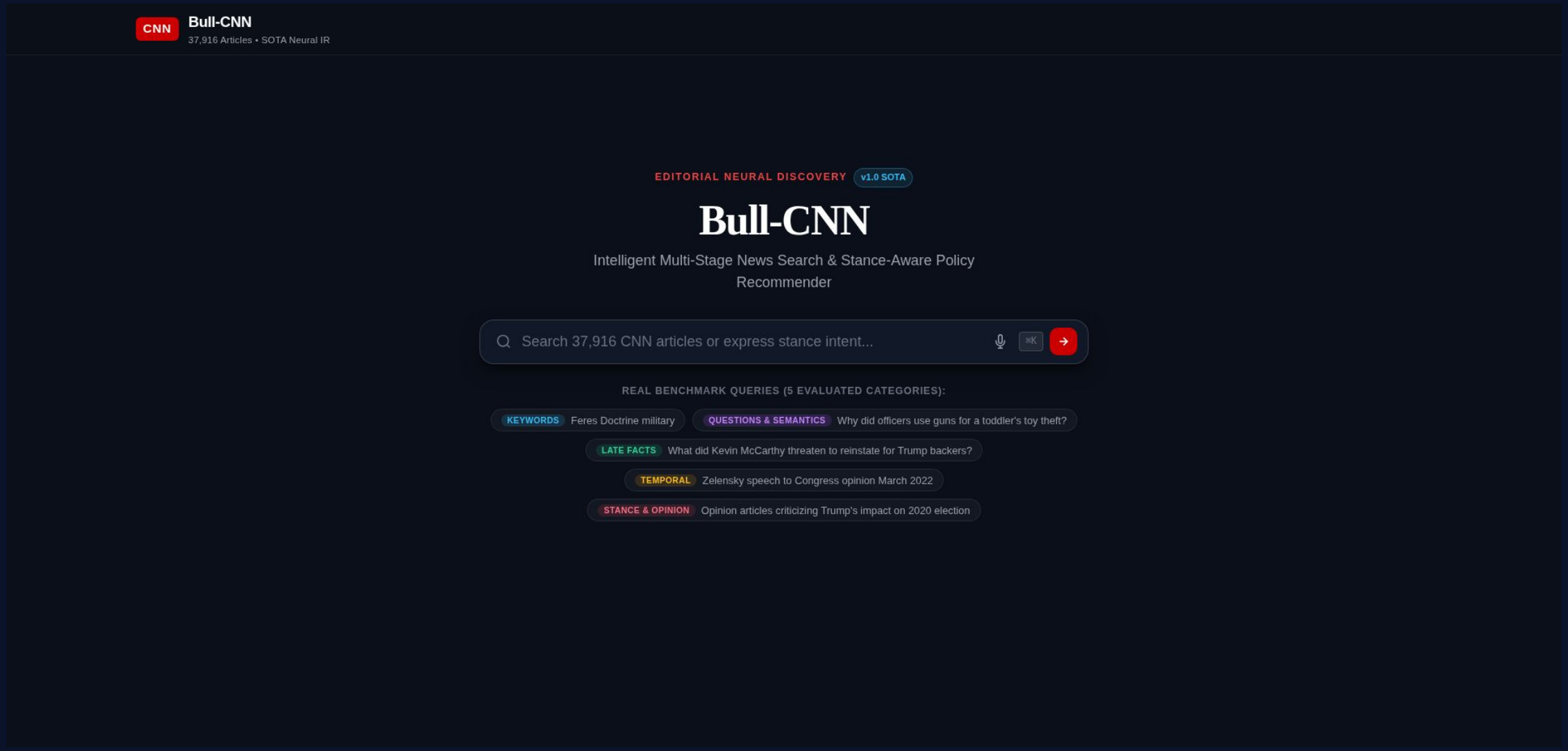
4. Dynamic Temporal Recency Priors

Limitation: Static 2011–2022 archive treats all timestamps equally unless explicitly filtered.

Outlook: Add query-adaptive exponential time-decay e^(-lambda * dt) for breaking news vs. archival research.

PART II • INTERACTIVE SPLIT-CANVAS WEB APPLICATION & FASTMCP SERVER

Live Demonstration: Search & Perspective Switcher



PART III • GOALS 2 + 3

From PoC to Production: Team, Roadmap & Architecture

Agile Process & Project Team Roles

Six-Sprint Implementation Roadmap

Sovereign BullSequana Production Architecture

Agile Sprint Governance & Project Team (3 Months)

Agile Delivery Framework

- Sprint Cadence: 6 two-week iterations
- Tooling: GitHub Projects (Kanban + Sprint Milestones), automated CI/CD pytest gates, PR code review & MLflow tracking

Ceremonies & Rhythm

- Daily Standup (15 min, daily): Fast sync across DS, DE & Fullstack to unblock pipeline & model dependencies
- Review, Planning & Refinement (1.5h, twice a week): Joint technical alignment, live metric review & backlog prioritization with customer Product Owner & editorial leads
- Monthly In-Person Retrospective & Milestone Demo (Every 4 Weeks — W4, W8, W12): On-site workshop reviewing sprint velocity, human gold QA agreement & live staging sign-off

Role	Capacity	Core Responsibilities
Senior Data Scientist	**1.0 FTE**	• Starts Human Gold Calibration early in Sprint 2 with editorial experts • Distills stance & 12-frame classifier into <15ms encoder head • Implements Counterfactual LTR & temporal recency priors
Senior Data Engineer	**1.0 FTE**	• Builds Apache Kafka event bus & CMS CDC connectors • Implements streaming BPE chunking & async enrichment workers • Manages PostgreSQL 16, HNSW/BM25F index & Redis ops
Fullstack Developer	**1.0 FTE**	• Starts Split-Canvas UI & CMS integration in Sprint 3 (Weeks 5–10) • Productionizes FastAPI backend, API Gateway & SSO/RBAC auth • Builds real-time clickstream & dwell-time telemetry emitter
Test Engineer (Optional)	**0.25–0.5 FTE (Part-Time)**	• Aligns architecture with customer IT & BullSequana AI sizing • Automated E2E regression suite, security & GDPR audit • Load/stress testing for <15ms autocomplete & <80ms search SLA

GOALS 2 + 3 • 3-MONTH EDITABLE GANTT ROADMAP (6 × 2-WEEK SPRINTS)

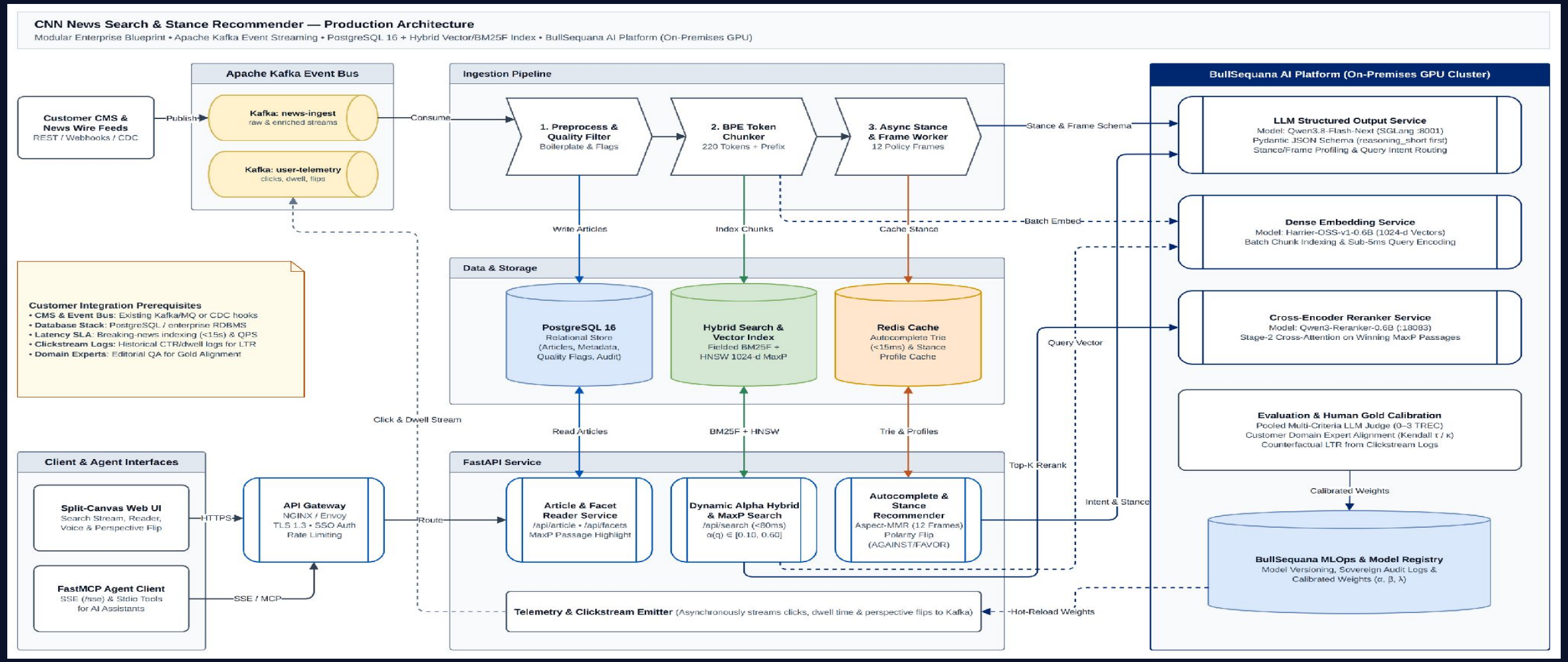
3-Month Engineering Roadmap (Weeks 1–12)

ENGINEERING WORKSTREAM / OWNER	SPRINT 1 Weeks 1–2	SPRINT 2 Weeks 3–4	SPRINT 3 Weeks 5–6	SPRINT 4 Weeks 7–8	SPRINT 5 Weeks 9–10	SPRINT 6 Weeks 11–12
1. Customer Discovery, CMS Schema, Logs (All)	W1–W2: Prereqs, CMS & SLA Spec					
2. Human Gold Calibration with Domain Experts (DS)		Sprints 2–4 (W3–W8): 200-Query Gold Set & Judge Calibration (κ ≥ 0.78)				
3. Apache Kafka Event Bus, PostgreSQL 16 (DE)		Sprints 2–3 (W3–W6): Kafka Ingest & PostgreSQL 16				
4. Hybrid BM25F + HNSW Index & BullSequana Setup (DE/DS)		Sprints 2–3 (W3–W6): Hybrid Index & BullSequana GPU				
5. Async Write-Time Stance Worker & Distillation (DS/DE)			Sprints 3–5 (W5–W10): <15ms Distilled Stance Head & LTR			
6. Fullstack Split-Canvas UI, CMS Integration & SSO (FS)			Sprints 3–5 (W5–W10): Split-Canvas UI, API Gateway & Click Telemetry			
7. SLA Load Testing (<80ms), Security Audit & Go-Live (All)						W11–W12: SLA & Go-Live
★ MONTHLY IN-PERSON RETRO & MILESTONES	★ End of Sprint 2 (Week 4): M1 In-Person Retro & Ingest Sign-Off		★ End of Sprint 4 (Week 8): M2 In-Person Retro & Gold QA Sign-Off		★ End of Sprint 6 (Week 12): M3 In-Person Retro & Production Go-Live	

GOAL 3 • PRODUCTION TARGET ARCHITECTURE (NATIVE DRAW.IO BLUEPRINT)

End-to-End Production Architecture with BullSequana AI

Apache Kafka ingest, PostgreSQL 16 + Hybrid BM25F/HNSW index, Redis cache & BullSequana GPU services



Modular Enterprise Blueprint: Apache Kafka Ingest • PostgreSQL 16 Single-Copy Store • Hybrid BM25F + HNSW Index • Sovereign BullSequana AI Platform

EXECUTIVE SUMMARY • KEY NUMBERS & VISUAL SYNTHESIS (PRE-CLOSE)

What We Built & Proved: At a Glance

37,916 Articles

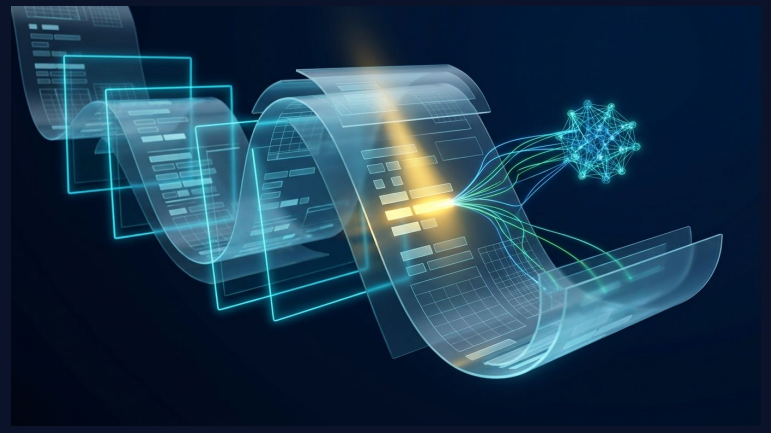
208,187 Contextual BPE Passages
Zero Truncation up to 23,670 Tokens

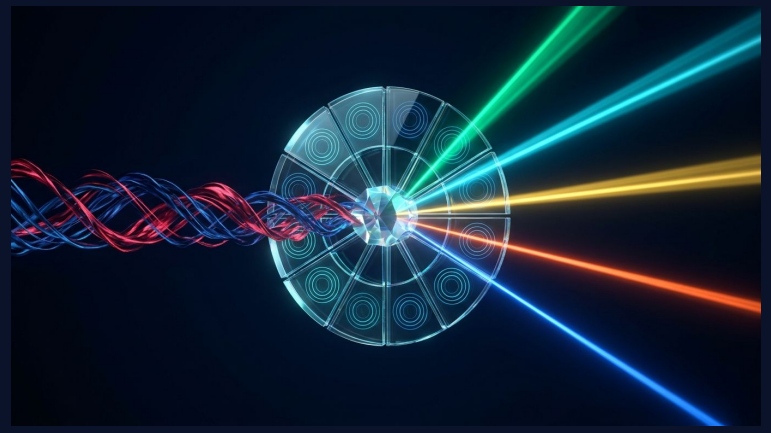
100% @ 5

Stance Precision@5 • 5/5 Policy Frames
Instant 1-Click Polarity Flip

0.825 nDCG@10

83.6% Strict Rec@5 • 250 Queries
8.8 ms Hybrid • 44.6 ms Cross-Encoder







1. Deep Hybrid Search & Reranking

- Fielded BM25F + 220-Token BPE Windows + Debiased MaxP
- Dynamic Alpha Hybrid + Qwen3 MaxP Passage Forwarding

2. 12-Frame Stance Recommender

- Constrained CoT Stance Profiling + Hard Stance Gate
- Aspect-MMR across 12 Policy Frames + 1-Click Flip

3. 3-Month Sovereign Rollout

- 6 Agile Sprints (DS + DE + Fullstack) & Early Gold QA
- Apache Kafka + PostgreSQL 16 + BullSequana AI GPU

“Search Finds Answers. Perspective Builds Understanding.”



Thank You! • Ready to Launch Sprint 1 — Or Ready to Hire? Let's Talk Q&A!

Live Demo: http://127.0.0.1:8000 | Architecture: bull_cnn_production_architecture.drawio | Sovereign BullSequana AI